JUAN PABLO ROJO

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linkedin.com/in/juan-pablo-rojo-83b843243 | github.com/juanpablorojo1 | Portfolio: jpr-web-mu.vercel.app
Cordoba, Argentina — On-site in Cordoba or remote worldwide

PROFESSIONAL PROFILE

Biomedical Engineer (UNC, 2026) specializing in artificial intelligence, computer vision, and medical image processing. Developed TraumaVision AI — a clinical decision support web app for pediatric wrist fracture detection using AI (internal test: sensitivity 96.7%, specificity 93.1%, AUC-ROC 0.987); thesis submitted, project now scaling as a startup. Founder of Eleven Estudio Digital. End-to-end profile: machine learning, computer vision, full-stack web development, and clinical biomedical engineering.

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas | deep learning, convolutional neural networks (CNN), transfer learning, fine-tuning | Claude API | medical image processing
Computer Vision: YOLOv8 (object detection, Ultralytics) | image classification, fracture detection, clinical radiological image analysis
Web Development & APIs: Next.js, React, FastAPI, SQLAlchemy, SQLite, Supabase | HTML/CSS/JS, GSAP, Vercel | REST API, AI model integration and deployment to production
Data & Visualization: Matplotlib, Pandas | interactive dashboards | automated PDF reports (ReportLab)
Biomedical Engineering: corrective and preventive maintenance of medical equipment | JCI, ISO/IEC, ANMAT standards | SaMD (Software as Medical Device) | healthcare regulatory compliance
Tools & Marketing: Python, Git/GitHub, Vercel, Supabase, Claude Code | Meta Ads, Google Ads, SEO | end-to-end project management

FEATURED PROJECTS

TraumaVision AI — Clinical Decision Support System 2025 – present
Senior Thesis (submitted) + Startup | National University of Cordoba

- Developed a web application for pediatric wrist fracture detection using AI, designed to assist trauma surgeons in reading X-ray images.
- Trained and validated an object detection model on GRAZPEDWRI-DX with a patient-level split. Internal test (3,038 images): sensitivity 96.7% | specificity 93.1% | AUC-ROC 0.987 | mAP@50 0.944. External test on PediURF without retraining: sensitivity 97.4% | specificity 88.3%. Retrospective evaluation; not registered with ANMAT and not cleared for clinical use.
- Built the web app around the model: FastAPI, SQLAlchemy, SQLite and Jinja2, with DICOM/ZIP study upload, bounding-box viewer, automated PDF reports (ReportLab), and per-user accounts.
- Now scaling as a startup through EmprendeU; regulatory framework analyzed as SaMD (Software as Medical Device) under ANMAT Disposition 64/2025.
- Stack: Python | PyTorch | YOLOv8m (Ultralytics) | OpenCV | FastAPI | SQLAlchemy | HTML/CSS/JS

AWARDS & RECOGNITION

1st Place — Rocking the Industry (World's First Industrial Hackathon) 2025
RTS Group | Team: Algarroba LID | 150 multidisciplinary teams

- Anonymous challenge on a remote nuclear power plant: diagnose operational failures and propose an AI-driven control system with human-in-the-loop agents for sensor supervision, under real time pressure.

- Designed a simple and robust architecture integrating data, processes, and operational control: immediate stabilization actions combined with long-term redesign strategies to make the plant more efficient, safe, and predictable.
- Integrated perspectives from design, renewable energy, sustainability, and engineering to deliver a decision-support system that anticipates problems and improves operational reliability in critical conditions.

WORK EXPERIENCE

AI & Automation Specialist Aug 2026 – present
Leex Medical | Cordoba, Argentina | leexmedical.com

- Updating and developing websites to improve SEO and GEO/AEO positioning (traditional search engines and Artificial Intelligence searches).
- Optimizing advertising campaigns to improve lead volume and quality: Google Ads and Meta.
- Automating the generation and updating of commercial documents (technical data sheets, brochures, among others) using AI.
- Generating and tracking sales calendars using AI.
- Automating tender/bidding workflows using AI.
- Automating content generation for social media, blogs, and newsletters using AI (a hybrid process in formats such as video, with human validation before publishing).
- Developing and monitoring AI agents that interact on social media and on brand websites (Leex/Kibbo) with the goal of generating leads.

AI Engineer Aug 2026 – present
Kibbo | Cordoba, Argentina | Kibbo.io

- Updating and developing websites to improve SEO and GEO/AEO positioning (traditional search engines and Artificial Intelligence searches).
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Founder — Web Development & AI Agents May 2026 – present
Eleven Estudio Digital | Cordoba, Argentina | eleven-pagina.vercel.app

- Founded and lead an AI-native digital studio: built a 38-agent coordinated pipeline (Claude API) covering discovery, strategy, branding, development, and QA — with 4 products live in production.
- Built client websites with Next.js, React, GSAP, and Vercel; integrated AI agents via Claude API for 24/7 automated customer service on WhatsApp and Instagram.

- Managed the full project lifecycle: discovery, strategy, design, development, delivery, and ongoing monthly maintenance.
- Ran Meta Ads and Google Ads campaigns and ongoing SEO strategies for clients in the services and retail sectors.

Biomedical Engineering Internship Mar – Jul 2024
Sanatorio Allende S.A. | Nueva Cordoba, Argentina

- Performed corrective and preventive maintenance on medical equipment in operating rooms, pediatric and adult ICUs, general wards, and sterilization units.
- Coordinated with medical and technical staff to diagnose equipment failures and restore critical device availability.
- Ensured compliance with Joint Commission International (JCI) standards across all maintained equipment and hospital processes.

Undergraduate Teaching Assistant — Human Physiology Aug 2023 – Jun 2024
FCEFyN, National University of Cordoba | Cordoba, Argentina

- Supported instruction of Human Physiology for Biomedical Engineering students.
- Translated complex physiological concepts into accessible language and led collaborative study groups.

Equipment Technician Dec 2023 – Mar 2024 | Dec 2024 – Mar 2025
Snow Summit LLC | Big Bear Lake, California, USA

- Developed professional-level English fluency operating in a 100% English-speaking, high-demand customer environment.
- Managed the full equipment acquisition process and performed technical maintenance under operational pressure.

EDUCATION

Biomedical Engineering 2020 – 2026 (expected)
National University of Cordoba — Faculty of Exact, Physical and Natural Sciences (FCEFyN)

- GPA: 7.45/10 with no failures | 90% degree completion
- Thesis: TraumaVision AI — Clinical Decision Support System based on Deep Learning for Fracture Detection in Pediatric Wrist Radiographs

ADDITIONAL TRAINING

Applied Data Science with Python Specialization Coursera
University of Michigan

- Data visualization, applied machine learning, network analysis, supervised learning. Libraries: NumPy, Pandas, Matplotlib, Scikit-learn.

LANGUAGES

- English: B2 Level — EF International Language Campus, Eastbourne, United Kingdom + 2 seasons of professional work in California, USA
- Spanish: Native